AMENDMENT NO. 4 JUNE 2011
TO
IS 277 : 2003 GALVANIZED STEEL SHEETS (PLAIN AND CORRUGATED) — SPECIFICATION

(Sixth Revision)

(*Page* 1, *clause* **5.1**) — Substitute the following for the existing:

'**5.1** The base metal chemistry of plain galvanized sheets and coils shall conform to 1S 1079 or IS 513 as the case may be.'

[*Page* 1, *clause* **6** (*see also Amendments No.* 1 *and* 2)] — Substitute the following for the existing table:

Table 1 Mechanical Properties
(*Clause* 6)

Sl No.	**Quality**		**Yield Stress** R_e MPa	**Tensile Strength** R_m MPa	**Elongation, Percent** *A Min*	
	Designation	Name			L_o = 80 mm	L_o = 50 mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	GP	Ordinary	—	—	—	—
ii)	GPH	Ordinary-Hard	400, *Min*	—	—	—
iii)	GC	Corrugated ordinary	—	—	—	—
iv)	GPL	Drawing (Lock Forming)	350, *Max*	450, *Max*	24	25
v)	GPD	Deep drawing	280, *Max*	430, *Max*	26	27
vi)	GPED	Extra deep drawing	260, *Max*	430, *Max*	28	31
vii)	GPIF	Interstitial free (Stabilized)	240, *Max*	370, *Max*	34	36

NOTES

1 Elongation values shall be reduced by 4 units for thickness values 0.5 mm and lower and shall be reduced by 2 units for thickness value above 0.5 mm up to 0.7 mm.

2 *Ordering condition* — Steel when ordered to mechanical properties shall, at the time the steel is made available for shipment, satisfy the applicable requirements of above table.

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(*Page* 2, *clause* **8**) — Substitute the following for the existing:

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'8 MECHANICAL PROPERTIES

8.1 Tensile Test

8.1.1 Tensile test shall be carried out only, if specified by the purchaser.

8.1.2 When specified, the tensile test shall be carried out in accordance with IS 1608 as applicable, and the values of tensile strength, yield stress and percentage elongation shall conform to the requirements specified in Table 1.

8.1.3 Tensile test values apply to transverse specimen in case of sheet/strips. Transverse test piece shall be taken midway between the centre and the edge of the sheet as-rolled.

8.1.4 The yield strength values apply to the 0.2 percent proof stress, if the yield strength is not clearly distinctive, otherwise the values apply to lower yield strength.

8.2 Bend Test

8.2.1 *Test Samples*

Bend test for the purpose of conformity shall be carried out at the rate of one set of 2 samples for every 1 000 plain sheets or part thereof. However, bend test shall not be applicable to GPH, GC and GCH grades intended for corrugation.

8.2.1.1 One bend test shall be conducted for every coil.

8.2.1.2 For bend test, the test piece shall be 230 mm long and 75 mm to 100 mm wide cut across the direction of rolling.

8.2.1.3 Specimens for bend tests shall be free from burrs. Filling or machining to remove burrs is permitted. Cracks of the base metal developing at the edge of the specimen or coarse grain developing at the line of the bend shall be disregarded.

8.3 Requirements

Samples of galvanized steel sheets selected as described in **8.2.1** shall withstand bending through 180° around a mandrel having diameter specified in Table 3

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without peeling or flaking of zinc coating. Crack or fracture of base metal, except those indicated in **8.2.1.3**, shall not be permitted.'

(*Page* 4, *clause* **14.1.3.1**) — Substitute the following for the existing

'Sheets of sizes and corrugation patterns other than those specified above may be supplied, if agreed to by the purchaser and manufacturer.'

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(MTD 4)

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